

Chemical Reactions and Equations

I] Textbook Questions:

★ 1 Mark Objectives: ★

1. Match the columns.

[2024, 2025]

Reactants	Products (Answers)	Type of Chemical Reaction (Answers)
$\text{BaCl}_2(\text{aq}) + \text{ZnSO}_4(\text{aq})$	$\text{BaSO}_4\downarrow + \text{ZnCl}_2(\text{aq})$	Double displacement
$2\text{AgCl}(\text{s})$	$2\text{Ag}(\text{s}) + \text{Cl}_2(\text{g})$	Decomposition
$\text{CuSO}_4(\text{aq}) + \text{Fe}(\text{s})$	$\text{FeSO}_4(\text{aq}) + \text{Cu}(\text{s})$	Displacement
$\text{H}_2\text{O}(\text{l}) + \text{CO}_2(\text{g})$	$\text{H}_2\text{CO}_3(\text{aq})$	Combination

2. Identify the endothermic and exothermic reaction.

[2020, 2022, 2025]

a. $\text{HCl} + \text{NaOH} \rightarrow \text{NaCl} + \text{H}_2\text{O} + \text{heat}$ — Exothermic reaction

b. $2\text{KClO}_3(\text{s}) \xrightarrow{\Delta} 2\text{KCl}(\text{s}) + 3\text{O}_2\uparrow$ — Endothermic reaction

c. $\text{CaO} + \text{H}_2\text{O} \rightarrow \text{Ca}(\text{OH})_2 + \text{heat}$ — Exothermic reaction

d. $\text{CaCO}_3(\text{s}) \xrightarrow{\Delta} \text{CaO}(\text{s}) + \text{CO}_2\uparrow$ — Endothermic reaction

3. Choose the correct option from the brackets and complete the sentences.

(oxidation, displacement, electrolysis, reduction, zinc, copper, double displacement, decomposition)

a. To prevent rusting, a layer of **zinc** metal is applied on iron sheets. [2022]

b. The conversion of ferrous sulphate to ferric sulphate is an **oxidation** reaction. [2020]

c. When electric current is passed through acidulated water, **electrolysis** of water takes place.

d. Addition of aqueous solution of ZnSO_4 to an aqueous solution of BaCl_2 is an example of **double displacement** reaction.

★ 2 Marks Questions: ★

Q. Give Reasons:

- 1. When the gas formed on heating limestone is passed through freshly prepared lime water, the lime water turns milky.** [2024]

Ans:

1. On heating, limestone (calcium carbonate) undergoes thermal decomposition to form calcium oxide (quick lime) and carbon dioxide.

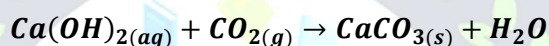


Calcium
Carbonate

Calcium
oxide

Carbon dioxide

2. When carbon dioxide is passed through lime water (calcium hydroxide) it turns milky due to formation of white precipitate of Calcium carbonate.



Calcium
hydroxide

Carbon
dioxide

Calcium
carbonate

- 2. It takes time for pieces of Shahabad tile to disappear in HCl, but its powder disappears rapidly.**

Ans:

- Shahabad tiles (CaCO_3) react with HCl to produce effervescence of CO_2 gas.
- Rate of chemical reaction depends on the size of the particles of the reactants.
- The smaller the size of the reactant particles, the higher the rate of reaction.
- Powdered Shahabad tile has smaller particles than pieces of Shahabad tiles. Hence, the powdered Shahabad tile disappears rapidly in HCl.

- 3. While preparing dilute sulphuric acid from concentrated sulphuric acid in the laboratory, the concentrated sulphuric acid is added slowly to water with constant stirring.**

Ans:

- In the process of dilution of concentrated sulphuric acid with water, very large amount of heat is liberated.
- As a result, water gets evaporated instantaneously, if it is poured in to the concentrated sulphuric acid, which may cause an accident.
- To avoid this, required amount of water is taken in a glass container and small quantity of concentrated sulphuric acid at a time is added with stirring.
- Therefore, only a small amount of heat is liberated at a time.

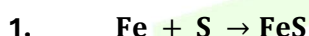
- 4. It is recommended to use air tight container for storing oil for long time.**

[2022]

Ans:

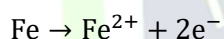
1. When we use old, left over cooking oil for making food stuff, it is found to have foul odour called rancidity.
2. If food is cooked in such oil, its taste also changes.
3. When oil or ghee is left aside for a long time or fried food is left aside for a long time it undergoes air oxidation and becomes rancid.
4. The process of oxidation reaction of food stuff can also be slowed down by storing it in air tight container. Therefore, it is recommended to use air tight container for storing oil for long time.

Q. Identify from the following reactions the reactants that undergo oxidation and reduction.

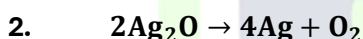
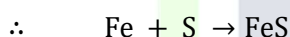
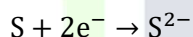


Ans. In this reaction, Fe undergoes oxidation and S undergoes reduction.

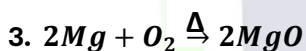
Fe loses 2 electrons



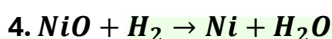
S gains 2 electrons



Ans. In this reaction, Ag_2O undergoes reduction as it loses oxygen.



Ans. In this reaction, Mg undergoes oxidation as it combines with oxygen.



Ans. In this reaction, NiO undergoes reduction and H_2 undergoes oxidation.

★ 3 Marks Questions: ★

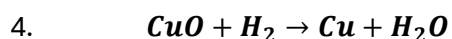
Q. Answer the following.

1. What is the reaction called when oxidation and reduction take place simultaneously? Explain with one example.

OR What is redox reaction? Explain with the help of balanced chemical equation. [2022]

Ans:

1. When oxidation and reduction take place simultaneously, the reaction is called Redox reaction.
2. Redox Reaction = Reduction + Oxidation
3. e.g. When hydrogen gas is passed over black copper oxide a reddish coloured layer of copper is formed.



5. In this reaction an oxygen atom goes away from CuO (copper oxide), which means that reduction of copper oxide takes place.
6. Hydrogen molecule takes up oxygen atom and water (H_2O) is formed meaning, oxidation of hydrogen takes place.

2. How can the rate of the chemical reaction, namely, decomposition of hydrogen peroxide be increased?

Ans:

1. The decomposition of hydrogen peroxide (H_2O_2) into water and oxygen takes place slowly at room temperature.
2. $2\text{H}_2\text{O}_2 \rightarrow 2\text{H}_2\text{O} + \text{O}_2$
3. The rate of this reaction can be increased by adding manganese dioxide (MnO_2) powder in hydrogen peroxide.
4. Here, MnO_2 acts as a catalyst which increases the rate of the chemical reaction. MnO_2 does not undergo any chemical change as it is used as a catalyst.

3. Explain the term reactant and product giving examples.

[2019]

Ans:

1. The substances taking part in chemical reaction are called reactants
2. whereas the substances formed as a result of a chemical reaction by formation of new bonds are called products.
3. Some examples are:
 - a. Burning of coal in air. The reaction can be written as –

$$\text{C} + \text{O}_2 \rightarrow \text{CO}_2$$

In this reaction, coal (carbon) and oxygen are reactants while carbon dioxide is the product.
 - b. $\text{CuSO}_4 + \text{Zn} \rightarrow \text{ZnSO}_4 + \text{Cu}$

In this reaction, CuSO_4 and Zn are the reactants and ZnSO_4 and Cu are the products.

4. Explain the similarity and difference in the two events, namely adding NaOH to water and adding CaO to water.

[2019]

Ans.

Similarity

1. When solid NaOH is added to water, heat is given out and hence the temperature increases. This is an exothermic process.
2. Also, when CaO is added to water, heat is given out; hence, it is an exothermic reaction.

Difference

1. When NaOH is added to water, NaOH only dissolves in water and forms an aqueous NaOH solution. It is a physical change as no new product is formed.
2. When CaO is added to water, a chemical reaction takes place giving out heat, forming calcium hydroxide. Hence, it is a chemical change.

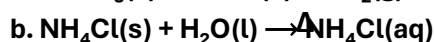
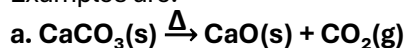
5. Explain the following terms with examples.

a. Endothermic reaction

Ans.

1. The chemical reaction in which heat is either absorbed from the surroundings or has to be supplied continuously from outside is called endothermic reaction.

2. Examples are:



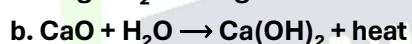
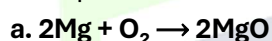
b. Combination reaction

[2023]

Ans.

1. When two or more reactants combine in a reaction to form a single product, it is called a combination reaction.

2. Examples are:

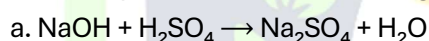


c. Balanced equation

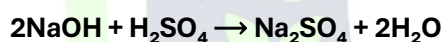
Ans.

1. In a chemical equation, if the number of atoms of the elements in the reactants is the same as the number of atoms of elements in the products. Such an equation is called a balanced equation.

2. For example, consider the following equations:



The balanced equation can be written as:



Thus, we can see that the number of atoms of elements in the reactants and products are equal. Hence, the equations are balanced.

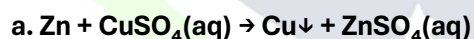
3. A chemical equation is incomplete and incorrect till it is balanced.

d. Displacement reaction

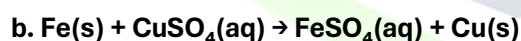
Ans.

1. The reaction in which the place of the ion of a less reactive element in a compound is taken by another more reactive element by formation of its own ions is called displacement reaction.

2. Examples are –



Here, Zn displaces Cu from CuSO_4 , as Zn is more reactive than Cu.



Here, the more reactive element Fe displaces the less reactive Cu from CuSO_4 and forms ferrous sulphate (FeSO_4).

★ 5 Marks Questions: ★

1. Observe the following picture and answer the questions.

OR

Observe the following picture and write down the chemical reaction with explanation

a. Which process is shown in the figure?

Ans: Rusting of iron is shown.

b. What is rust?

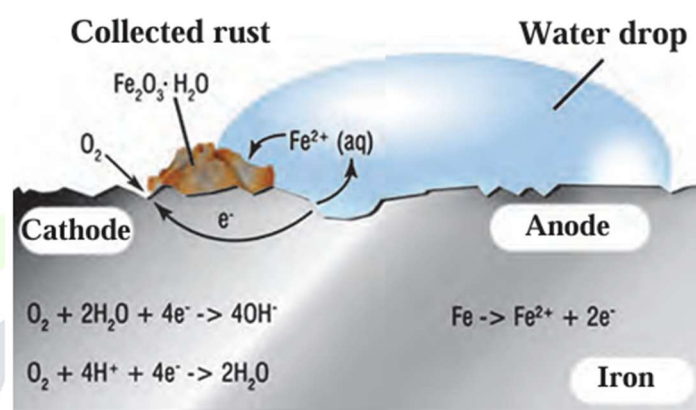
[2020]

Ans: When iron is exposed to air and moisture, a reddish coloured solid layer of hydrated ferric oxide is formed on its surface. This is called rust.

c. Write the chemical formula of rust.

[2020]

Ans: The chemical formula of rust is $\text{Fe}_2\text{O}_3 \cdot x\text{H}_2\text{O}$.

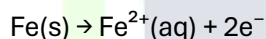


d. Explain the chemical reaction shown in the figure. Write the reactions on anode and cathode.

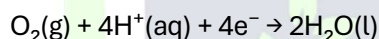
[2020]

Ans:

Fe is oxidized to Fe^{2+} in the anode region.

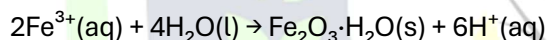


O_2 is reduced to form water in the cathode region.



When Fe^{2+} ions migrate to the anode region, get oxidized to form Fe^{3+} ions.

Rust formed collects on the surface of iron.



e. What is corrosion?

[2020]

Ans: Due to various components of atmosphere, oxidation of metals takes place, resulting in their damage. This is called corrosion.

II] Extra - Previous Years' Questions: (PYQs)

★ 1 Mark Objectives: ★

MCQs:

- When the positive charge on an atom or an ion increases or the negative charge on them decreases it is called _____. [2025]
(A) Reduction (B) Oxidation (C) Displacement (D) Decomposition
- The reaction in which ions in the reactants are exchanged to form a precipitate is called as _____ reaction. [2025]
(A) Combination (B) Displacement (C) Decomposition (D) Double displacement
- The left-hand side of a chemical reaction represents _____. [2023]
(A) Product (B) Reactants (C) Catalyst (D) Indicator

True or False:

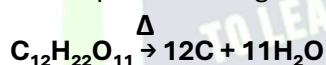
4. A redox reaction takes place during cellular respiration. (True) [2025]
5. Rancidity is oxidation process. (True) [2023]
6. Write the molecular formula of calcium carbonate. Ans: CaCO_3 [2024]
7. Correlation: CuI_2 : Brown :: AgCl : White [2023]

★ 2 Marks Questions: ★

1. What is mean by decomposition reaction? Write the chemical reaction of decomposition of sugar on heating. [2023, 2025]

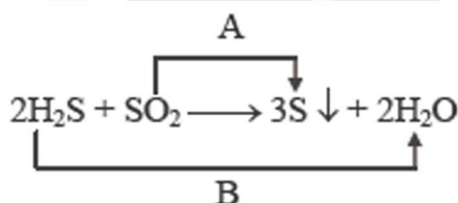
Ans:

1. The chemical reaction in which two or more products are formed from a single reactant is called decomposition reaction.
2. Decomposition of sugar:



Sugar

carbon



2. [2024]

(a) Identify the above chemical reaction.

Ans: The reaction is a Redox Reaction

(b) Name the type of reaction in A and B.

Ans: A: Reduction reaction B: Oxidation reaction

3. Distinguish between oxidation and reduction. [2026]

Oxidation reaction	Reduction reaction
(i) It is a chemical process in which a substance gains oxygen or loses hydrogen.	(i) It is a chemical process in which a substance gains hydrogen or loses oxygen.
(ii) During oxidation reaction, an atom or an ion loses electron.	(ii) During reduction reaction, an atom or ion gains electron.
(iii) Example: $\text{C} + \text{O}_2 \rightarrow \text{CO}_2 \uparrow$	(iii) Example: $\text{C} + 2\text{H}_2 \rightarrow \text{CH}_4$



State whether the above chemical reaction is balanced or not. From the number of atoms present in the reactants and products, fill in the blanks: [2026]

Elements	Reactants (Left Side) No. of atoms	Products (Right Side) No. of atoms
Na	2	2
O	6	5
H	4	2
S	1	1

Therefore, the chemical equation is not balanced.

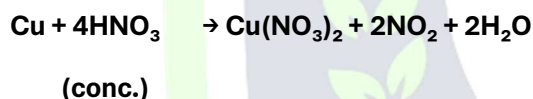
★ 3 Marks Questions: ★

1. a. What happens when copper reacts with concentrated nitric acid? What is the colour of the gas released during the reaction? [2025]

Ans: When copper (Cu) reacts with concentrated nitric acid (HNO₃), it undergoes an oxidation-reduction reaction forming copper nitrate and a reddish-brown poisonous nitrogen dioxide gas with water. The gas released is reddish-brown in colour.

b. Write its balanced chemical equation.

Ans:



c. Write the names of reactants and products.

Ans:

Reactants: Copper and concentrated nitric acid.

Products: Copper nitrate, nitrogen dioxide and water.

2. What information will you get from the given reaction? [2024]



Ans:

1. **Reactants:** Calcium carbonate

Products: Calcium Oxide and Carbon dioxide

2. On heating, limestone (calcium carbonate) undergoes thermal decomposition to form calcium oxide (quick lime) and carbon dioxide.

3. It is an Endothermic and Decomposition reaction.

3. a. What are the factors affecting the rate of chemical reaction: [2024]

Ans: The factors affecting the rate of chemical reaction are:

1. Nature of the Reactants
2. Size of the Particles of Reactants
3. Concentration of the reactants
4. Temperature of the Reaction
5. Catalyst

b. Explain anyone factor:

Temperature of the Reaction:

1. While studying decomposition reaction, we have carried out decomposition of lime stone.
2. In this activity, the lime water does not turn milky before heating the lime stone; because of the zero rate of reaction.
3. From this it is learnt that the rate of a reaction increases on increasing the temperature.

4. With reference to the given diagram, answer the following questions: [2023]

a. Give type of chemical reaction.

Ans: Decomposition reaction

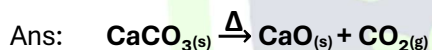
b. Give the names of reactants and products.

Ans:

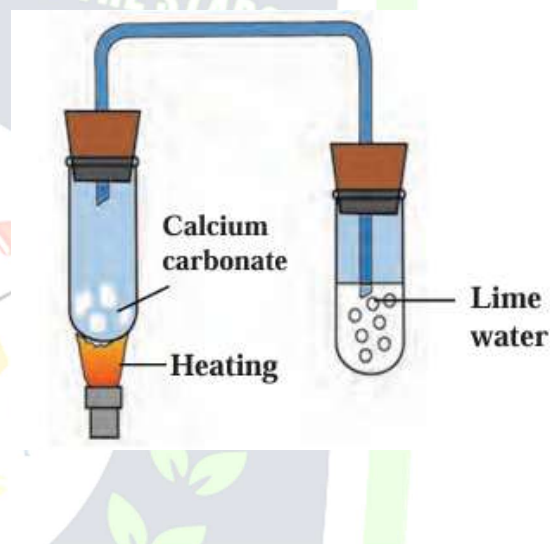
Reactant: Calcium Carbonate (CaCO_3)

Product: Calcium oxide (CaO) and Carbon dioxide (CO_2)

c. Write down the balanced chemical equation.



Calcium	Calcium	Carbon
Carbonate	Oxide	dioxide



5. Identify the physical and chemical changes from the following Phenomena: [2026, 2023]

- Transformation of ice into water. **(Physical)**
- Ripening of fruit. **(Chemical)**
- Milk turned into curd. **(Chemical)**
- Evaporation of water. **(Physical)**
- Digestion of food in the stomach. **(Chemical)**
- Iron fillings get attracted towards the magnet. **(Physical)**